

# A Macroeconomic Perspective on Taxing Multinational Enterprises

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# Base Erosion and Profit Shifting (BEPS)

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Tax planning strategies of **Multinational Enterprises (MNE)** that exploit gaps in tax rules to:

- artificially shift profits to low or no-tax locations with little or no economic activity
- erode tax bases through deductible payments such as interest or royalties

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The magnitude of BEPS:

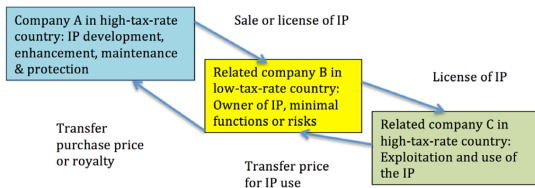
- **OECD estimate: \$100-\$240 billion** global revenues annually, **4-10%** of global corporate tax revenues
- **Tørsløv et al. (2022): 36%** of multinational profits were shifted to tax havens globally in 2015
- **Guvenen et al. (2022): 37%** of income recorded by US multinationals' foreign affiliates was shifted out of the US

# Multinationals shift profits by reallocating IP rights

Two ways of reallocation to the affiliates in low-tax jurisdictions:

- **IP sale**: parent loans affiliate money to buy IP outright
- **Cross licensing**: affiliate pays for portion of parent's R&D expenses

Reallocation often occurs at price **below IP's "market value"**, violating **arm's length principle**.



Source: Neubig and Wunsh-Vincent (2017)

# OECD Two-Pillar Proposal on Taxing MNEs

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The largest tax agreement in history to address BEPS problem:

- signed in October 2021 by 137 countries and tax jurisdictions
- signatories account for **90% of global GDP**
- to be implemented in 2023

Two-Pillar Solution:

1. **Pillar One**: Sales profit allocation.
  - **25%** of profits above a set profit margin (residual profits) would be reallocated to the market jurisdictions where the MNE's **users and customers** are located
2. **Pillar Two**: Global minimum tax.
  - minimum effective corporate profits tax rate of **15%**

# What we do

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1. **Analytically characterize** the impact of transfer pricing and profit shifting on production inputs in simplified model.
2. Develop a general-equilibrium **macroeconomic framework** that reflects the key features of the current international tax regime.
3. **Quantify** the effects of Two-Pillar Solution proposed by the OECD.

# What we do

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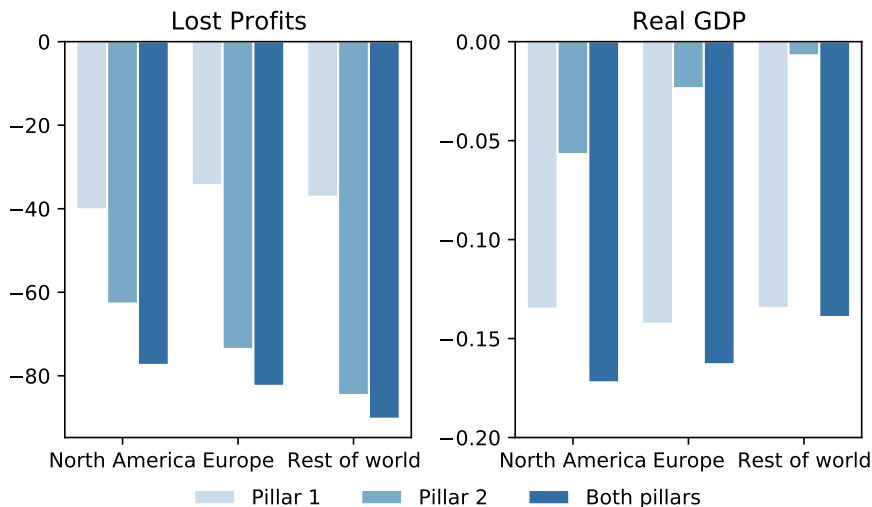
1. **Analytically characterize** the impact of transfer pricing and profit shifting on production inputs in simplified model.
2. Develop a general-equilibrium **macroeconomic framework** that reflects the key features of the current international tax regime.
3. **Quantify** the effects of Two-Pillar Solution proposed by the OECD.

## Findings:

1. **Key trade-off**: profit shifting erodes high-tax countries' tax bases, but incentivizes the MNEs to invest in the intangible capital.
2. **Main takeaway**: adverse macroeconomic impact of the landmark OECD policy designed to eliminate profit shifting.

# Main Results of the Paper

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# Outline of the Talk

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1. Theory of profit shifting and intangible investment.
2. Quantitative model.
3. Taking the model to the data.
4. Inspecting the economic mechanism.
5. The effects of OECD's Two-Pillar reform.

# THEORY OF PROFIT SHIFTING AND INTANGIBLES

# Environment

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- MNE with its parent division in  $i$  operates in  $K$  locations.
- Location  $k \in \{1, \dots, K\}$ :
  - Population:  $N_k$
  - Productivity:  $A_k$
  - Corporate profit tax rate:  $\tau_k$
  - Prices:  $p_k, w_k$
- Technology:

$$F(z, l_k) = A_k (N_k \mathbf{z})^\phi l_k^\gamma$$

where:

- $\mathbf{z}$  is **non-rival**, intangible capital
- $l_k$  is labor input
- DRS:  $(\gamma + \phi) < 1$

# Accounting profits

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**Free Transfer (FT):**  $z$  transferred at no cost across locations:

$$\begin{aligned}\pi_i &= p_i \left( A_i (N_i z)^\phi \ell_i^\gamma \right) - w_i l_i - p_i z \\ \pi_k &= p_k \left( A_k (N_k z)^\phi \ell_k^\gamma \right) - w_k l_k, \quad \forall k \neq i\end{aligned}$$

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**Transfer pricing (TP):** parent division retains legal ownership of  $z$  and licenses the rights to use it to its foreign affiliates.

$$\begin{aligned}\pi_i^{TP} &= \pi_i + \sum_{k \neq i} q_k z \\ \pi_k^{TP} &= \pi_k - q_k z \quad \forall k \neq i\end{aligned}$$

where

$$q_k \equiv \underbrace{\phi p_k N_k \left( A_k (N_k z)^\phi \ell_k^\gamma \right)}_{\text{Marginal Revenue Product of } z}$$

# Accounting profits

---

## Profit Shifting (PS):

$$\pi_i^{PS} = \pi_i + z \left[ \varphi \lambda \sum_k q_k - \lambda q_i + (1 - \lambda) \sum_{k \neq i} q_k - \mathcal{C}(\lambda) \sum_k q_k \right]$$

$$\pi_{i^*}^{PS} = \pi_{i^*} + z \left[ \lambda \sum_{k \neq i^*} q_k - (1 - \lambda) q_{i^*} - \varphi \lambda \sum_k q_k \right]$$

$$\pi_k^{PS} = \pi_k - q_k z \quad \forall k \neq i, i^*$$

where

- $\lambda \in [0, 1]$  a fraction of intangible capital  $z$  transferred to the tax haven
- $\mathcal{C}(\lambda)$  is the cost of shifting the fraction  $\lambda$
- $\varphi \leq 1$  is a markdown below the competitive price of  $z$
- $i^*$  is the tax haven i.e. :  $\tau_{i^*} = \min \{\tau_1, \dots, \tau_K\}$

# Profit maximization

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Consider after-tax MNE's profit maximization problem for each case:

$$\Pi^j \equiv \max_{z, \{l_k\}_{k=1}^K} \sum_{k=1}^K (1 - \tau_k) \pi_k^j$$

where

- $j \in \{FT, TP, PS\}$
- denote allocations of intangible capital accordingly by:  $z^{FT}, z^{TP}, z^{PS}$
- For  $j = PS$ , MNE chooses also  $\lambda$

# Profit-shifting margin

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## Assumption

Let  $\mathcal{C}(\lambda) \equiv \lambda - (1 - \lambda) \log(1 - \lambda)$ , implying  $\mathcal{C}'(\lambda) = -\log(1 - \lambda)$ ,  $\mathcal{C}(0) = 0$ ,  $\mathcal{C}(1) = 1$ , and  $\lambda \in [0, 1]$ .

The share of shifted intangible capital:

$$\lambda = 1 - \exp\left(-\frac{(1 - \varphi)(\tau_i - \tau_{i^*})}{1 - \tau_i}\right)$$

## Lemma

Under the Assumption, the share of shifted intangible capital  $\lambda$  is:

1. Decreasing in  $\varphi$ .
2. Decreasing in  $\tau_{i^*}$  with elasticity given by

$$\varepsilon_{\tau_{i^*}}^{\lambda} = -\frac{1 - \lambda}{\lambda} \left( \frac{1 - \varphi}{1 - \tau_i} \right) \tau_{i^*}$$



# Key trade-off: base erosion $\rightarrow$ increase in $z$

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## Proposition

*The following hold:*

1. *If  $\tau_i = \max\{\tau_k\}_{k=1}^K$  then  $z^{TP} < z^{FT}$ .*
2.  *$z^{PS} > z^{TP} \iff \varphi < 1$  and  $z_{PS} = z_{TP} \iff \varphi = 1$ .*
3.  *$z^{PS}$  is decreasing in  $\varphi$ .*
4.  *$z^{PS}$  is decreasing in  $\tau_{i^*}$ .*

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3.  $z^{PS}$  is decreasing in  $\varphi$ .
4.  $z^{PS}$  is decreasing in  $\tau_{i^*}$ .

We show

$$z^{TP} = \left( \frac{\sum_{k=1}^K \phi \Lambda_k}{p_i} \right)^{\frac{1-\gamma}{1-\phi-\gamma}} < \left( \frac{\sum_{k=1}^K (1-\tau_k) \phi \Lambda_k}{(1-\tau_i) p_i} \right)^{\frac{1-\gamma}{1-\phi-\gamma}} = z^{FT}$$

where  $\Lambda_k$  is a function of  $A_k, p_k, N_k, w_k$ . Then  $z^{PS}$  is

$$z^{PS} = z^{TP} \left( (1 - \mathcal{C}(\lambda)) + \frac{\lambda(1-\varphi)(\tau_i - \tau_{i^*})}{(1-\tau_i)} \right)^{\frac{1-\gamma}{1-\phi-\gamma}}$$

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3.  $z^{PS}$  is decreasing in  $\varphi$ .
4.  $z^{PS}$  is decreasing in  $\tau_{i^*}$ .

with the following elasticities:

$$\varepsilon_{\tau_{i^*}}^{z^{TP}} = 0$$

and

$$\varepsilon_{\tau_{i^*}}^{z^{PS}} = \frac{1 - \gamma}{1 - \phi + \gamma} \left( \frac{-\tau_{i^*}}{\tau_i - \tau_{i^*}} \right) \frac{1}{\left[ 1 + \frac{1 - C(\lambda)}{C'(\lambda)} \right]} < 0$$

# Pillar 1: profit allocation rule

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The MNE's tax base in jurisdiction  $k$  as:

$$T_k = \underbrace{\pi_k^r}_{\text{Routine profit}} + (1 - \theta) \times \underbrace{\pi_k^R}_{\text{Residual profit}} + \theta \times \underbrace{\frac{p_k y_k}{\sum_k p_k y_k}}_{\text{Sales share of } k} \times \underbrace{\Pi^R}_{\text{Global residual profit}}$$

where:

- $\pi_k^r = \mu p_k y_k$
- $\pi_k^R = \pi_k^{PS} - \pi_k^r$
- $\Pi^R = \sum_k \pi_k^R$

with two policy parameters:

- $\mu$  is the routine profit margin
- $\theta$  is the fraction of global residual profits allocated to jurisdiction  $k$  according to the sales share

# Pillar 1: profit allocation rule

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## Proposition

*The following hold:*

1.  $\hat{\lambda} < \lambda$  and  $\hat{z}^{PS} < z^{PS}$ .
2.  $\hat{\lambda}$  and  $\hat{z}^{PS}$  are decreasing in  $\theta$ .
3. The economy is less responsive to changes in  $\tau_i^*$ :

$$\left| \epsilon_{\tau_i^*}^{\hat{z}^{PS}} \right| < \left| \epsilon_{\tau_i^*}^{z^{PS}} \right|$$

$$\lambda = 1 - \exp \left( - \frac{(1 - \varphi)(\tau_i - \tau_{i^*})}{1 - \tau_i} \right)$$

# Pillar 1: profit allocation rule

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$$\left| \epsilon_{\tau_{i^*}}^{\hat{z}^{PS}} \right| < \left| \epsilon_{\tau_{i^*}}^{z^{PS}} \right|$$

$$\hat{\lambda} = 1 - \exp \left( - \frac{(1 - \varphi)(1 - \theta)(\tau_i - \tau_{i^*})}{1 - ((1 - \theta)\tau_i + \theta\hat{\tau})} \right).$$

where

$$\hat{\tau} \equiv \sum_i \tau_i \cdot \frac{p_i y_i}{\sum_k p_k y_k}.$$

# QUANTITATIVE MODEL

# Model environment

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- **Helpman, Melitz, and Yeaple (2004)** meets **McGrattan and Prescott (2010)** + **transfer pricing** and **profit shifting**
- $I$  productive regions
  - Representative consumer, gov't, and measure of firms
  - Differ in size, TFP, trade/FDI openness, corporate taxes
- 1 unproductive region (“tax haven”)
  - Gov't earns revenue by taxing profits of foreign MNEs' affiliates



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  - Differ in size, TFP, trade/FDI openness, corporate taxes
- 1 unproductive region (“tax haven”)
  - Gov't earns revenue by taxing profits of foreign MNEs' affiliates
- Firms in productive regions:
  - heterogeneous in productivity, compete monopolistically à la Melitz
  - **choose** whether to export and/or operate foreign affiliates
  - invest in **nonrival intangible capital** in parent division, charge foreign affiliates for rights to use it
  - **shift profits** to lowest-tax productive region and/or tax haven

# Firm's problem

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Each firm  $\omega$  in region  $i$  chooses:

- markets:
  - export destinations  $J_X$ , subject to fixed cost  $\kappa_{ij}^X$ . algebra
  - foreign affiliates  $J_F$ , subject to fixed cost  $\kappa_{ij}^F$ . algebra
- R&D and employment:
  - intangible capital investment  $z$ . algebra
  - local factors  $\ell_j$ . algebra
- profit shifting:
  - the share of intangible capital  $\lambda$  to shift algebra

to maximize after-tax global profit:

$$\max_{J_X, J_F, z, \lambda, \ell} \left\{ (1 - \tau_i) \left[ \pi_i^{PS}(\omega) - \sum_{j \in J_X} W_i \kappa_{ij}^X - \sum_{j \in J_F} W_i \kappa_{ij}^F \right] + \sum_{j \in J_F} (1 - \tau_j) \pi_{ij}^{PS}(\omega) \right\}$$

# Shifted Profits in the Model

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- Shifted profits by firm  $\omega$  from region  $j$

$$\tilde{\pi}_{ij}(\omega) = \pi_{ij}^{TP}(\omega) - \pi_{ij}^{PS}(\omega).$$

where:

- $\pi_{ij}^{PS}(\omega)$ : profit booked in region  $j$  by firm  $\omega$  based in region  $i$
  - $\pi_{ij}^{TP}(\omega)$ : the same object for TP scenario
- **Total shifted profits** from region  $j$  is

$$\tilde{\Pi}_j = \sum_{i=1}^I \int_{\omega \in \Omega_i, j \in J_F(\omega)} \tilde{\pi}_{ij}(\omega) d\omega.$$

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These measures can be defined in GE or PE:

- **PE**: hold fixed allocations of factors and measure how profitable each division would be if it was not allowed to profit-shift.

TAKING THE MODEL TO THE DATA

# Calibration

---

- **Five regions:** North America (NA), Europe (EU), Rest of the World (RoW) and:
  - Low Tax (LT): Belgium, Switzerland, Netherlands, Ireland etc.
  - Tax Haven (TH): Antigua, Aruba, the Bahamas, Barbados etc.

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- Profit shifting: match lost profit data from **Tørsløv et al. (2022)**.
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  - **Lost profit/GDP:** 0.6% for NA, 1.4% for EU, 0.7% for RoW and 9.5% for LT.
- We calibrate
  - **TFP** ( $A_i$ ) and **prod. dispersion** ( $\sigma_a$ ): GDP and firm size dist.
  - **Trade costs** ( $\kappa^X, \xi$ ): export participation and trade data.
  - **FDI costs** ( $\kappa^F, \sigma$ ): Domestic MNEs' and foreign MNEs' VA shares



## External Validation: semi-elasticity of profit shifting

---

**Semi-elasticity:** the percentage change of reported profit in response to a one percentage point change in the tax differential.

$$\log \pi_i^{k,PS}(\omega) = \beta_0 + \beta_\ell \log \ell_i^k(\omega) + \beta_z \log z^k(\omega) - \beta_\tau \hat{\tau}_i^k + \epsilon_i^k(\omega)$$

- $k$ : the index of the counterfactual economy
- $\hat{\tau}_i^k$ : tax differential between an MNE's home region and LT or TH.

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| Study                          | Data source                     | Headline point estimate |
|--------------------------------|---------------------------------|-------------------------|
| Johansson et al. (2017)        | ORBIS, 2000-2010                | 1.11                    |
| Heckemeyer and Overesch (2017) | Meta: 27 studies, 203 estimates | 0.79                    |
| Beer et al. (2020)             | Meta: 38 studies, 402 estimates | 0.98                    |
| This paper                     | Simulated model data            | 0.87                    |

# QUANTITATIVE EXPERIMENTS

# Inspecting the Mechanism: North America

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Free Transferring (FT)  $\rightarrow$  Transfer Pricing (TP)

- **On impact:**
  - **Domestic MNEs:** after-tax marginal revenue product  $z \downarrow \rightarrow$  intangible capital  $z \downarrow \rightarrow$  Value added (VA)  $\downarrow$
  - **Non-MNEs:** no direct effect
  - **Fiscal effect:** corporate tax base  $\uparrow/\downarrow$

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- **GE:**
  - **Reallocation effect:** wages  $\downarrow \rightarrow$  non-MNEs:  $z$  and VA  $\uparrow$
  - **FDI effect:** foreign-MNEs  $z$  and VA  $\uparrow$
  - **Fiscal effect:** corporate tax base  $\uparrow/\downarrow$
- **Total:**
  - **Macro and Fiscal Effects:** composition of forces

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Transfer Pricing (TP)  $\rightarrow$  Profit Shifting (PS): opposite direction

# Inspecting the Mechanism: Macro Effects

| Region                                                                       | Lost profits<br>(% GDP) | Corp. tax<br>rev. (% chg.) | Value added<br>(% chg.) | Tech. capital (% chg.) |              |                  |
|------------------------------------------------------------------------------|-------------------------|----------------------------|-------------------------|------------------------|--------------|------------------|
|                                                                              |                         |                            |                         | Total                  | Non<br>MNEs  | Domestic<br>MNEs |
| <i>(a) Effects of transfer pricing (no transfer pricing vs. no shifting)</i> |                         |                            |                         |                        |              |                  |
| North America                                                                | 0.00                    | 4.32                       | -0.16                   | <b>-0.54</b>           | <b>0.58</b>  | <b>-1.34</b>     |
| Europe                                                                       | 0.00                    | -2.34                      | -0.17                   | 0.12                   | 0.06         | 0.17             |
| Low tax                                                                      | 0.00                    | -2.17                      | -0.25                   | <b>0.74</b>            | <b>-0.75</b> | <b>2.28</b>      |
| Rest of world                                                                | 0.00                    | -0.41                      | -0.18                   | 0.05                   | 0.00         | 0.08             |
| <i>(b) Effects of profit shifting (no shifting vs. baseline)</i>             |                         |                            |                         |                        |              |                  |
| North America                                                                | 0.68                    | -3.82                      | 0.08                    | 0.21                   | -0.11        | 0.45             |
| Europe                                                                       | 1.05                    | -5.43                      | 0.06                    | 0.26                   | -0.07        | 0.55             |
| Low tax                                                                      | -4.37                   | 23.52                      | -0.04                   | -0.55                  | -0.60        | -0.49            |
| Rest of world                                                                | 0.50                    | -2.59                      | 0.04                    | 0.12                   | -0.06        | 0.27             |

# Inspecting the Mechanism: Macro Effects

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# Inspecting the Mechanism: VA decomposition

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| Region                                                                       | Value added (% chg.) |              |               |              |
|------------------------------------------------------------------------------|----------------------|--------------|---------------|--------------|
|                                                                              | Total                | Non MNEs     | Domestic MNEs | Foreign MNEs |
| <i>(a) Effects of transfer pricing (no transfer pricing vs. no shifting)</i> |                      |              |               |              |
| North America                                                                | <b>-0.16</b>         | <b>0.36</b>  | <b>-0.85</b>  | <b>0.35</b>  |
| Europe                                                                       | -0.17                | -0.15        | -0.11         | -0.31        |
| Low tax                                                                      | <b>-0.25</b>         | <b>-0.72</b> | <b>1.10</b>   | <b>-0.56</b> |
| Rest of world                                                                | -0.18                | -0.18        | -0.15         | -0.31        |
| <i>(b) Effects of profit shifting (no shifting vs. baseline)</i>             |                      |              |               |              |
| North America                                                                | 0.08                 | -0.00        | 0.15          | 0.15         |
| Europe                                                                       | 0.06                 | -0.02        | 0.17          | 0.04         |
| Low tax                                                                      | -0.04                | -0.33        | -0.29         | 0.64         |
| Rest of world                                                                | 0.04                 | -0.01        | 0.08          | 0.08         |

# Inspecting the Mechanism: Macro Effects

| Region                                                                       | Lost profits<br>(% GDP) | Corp. tax<br>rev. (% chg.) | Value added<br>(% chg.) | Tech. capital (% chg.) |              |                  |
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| <i>(a) Effects of transfer pricing (no transfer pricing vs. no shifting)</i> |                         |                            |                         |                        |              |                  |
| North America                                                                | 0.00                    | 4.32                       | -0.16                   | -0.54                  | 0.58         | -1.34            |
| Europe                                                                       | 0.00                    | -2.34                      | -0.17                   | 0.12                   | 0.06         | 0.17             |
| Low tax                                                                      | 0.00                    | -2.17                      | -0.25                   | 0.74                   | -0.75        | 2.28             |
| Rest of world                                                                | 0.00                    | -0.41                      | -0.18                   | 0.05                   | 0.00         | 0.08             |
| <i>(b) Effects of profit shifting (no shifting vs. baseline)</i>             |                         |                            |                         |                        |              |                  |
| North America                                                                | 0.68                    | -3.82                      | 0.08                    | <b>0.21</b>            | <b>-0.11</b> | <b>0.45</b>      |
| Europe                                                                       | 1.05                    | -5.43                      | 0.06                    | 0.26                   | -0.07        | 0.55             |
| Low tax                                                                      | -4.37                   | 23.52                      | -0.04                   | <b>-0.55</b>           | <b>-0.60</b> | <b>-0.49</b>     |
| Rest of world                                                                | 0.50                    | -2.59                      | 0.04                    | 0.12                   | -0.06        | 0.27             |

# Inspecting the Mechanism: Macro Effects

| Region                                                                       | Lost profits<br>(% GDP) | Corp. tax<br>rev. (% chg.) | Value added<br>(% chg.) | Tech. capital (% chg.) |             |                  |
|------------------------------------------------------------------------------|-------------------------|----------------------------|-------------------------|------------------------|-------------|------------------|
|                                                                              |                         |                            |                         | Total                  | Non<br>MNEs | Domestic<br>MNEs |
| <i>(a) Effects of transfer pricing (no transfer pricing vs. no shifting)</i> |                         |                            |                         |                        |             |                  |
| North America                                                                | 0.00                    | 4.32                       | -0.16                   | -0.54                  | 0.58        | -1.34            |
| Europe                                                                       | 0.00                    | -2.34                      | -0.17                   | 0.12                   | 0.06        | 0.17             |
| Low tax                                                                      | 0.00                    | -2.17                      | -0.25                   | 0.74                   | -0.75       | 2.28             |
| Rest of world                                                                | 0.00                    | -0.41                      | -0.18                   | 0.05                   | 0.00        | 0.08             |
| <i>(b) Effects of profit shifting (no shifting vs. baseline)</i>             |                         |                            |                         |                        |             |                  |
| North America                                                                | <b>0.68</b>             | <b>-3.82</b>               | <b>0.08</b>             | 0.21                   | -0.11       | 0.45             |
| Europe                                                                       | 1.05                    | -5.43                      | 0.06                    | 0.26                   | -0.07       | 0.55             |
| Low tax                                                                      | <b>-4.37</b>            | <b>23.52</b>               | <b>-0.04</b>            | -0.55                  | -0.60       | -0.49            |
| Rest of world                                                                | 0.50                    | -2.59                      | 0.04                    | 0.12                   | -0.06       | 0.27             |

# Inspecting the Mechanism: VA decomposition

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| Region                                                                       | Value added (% chg.) |              |               |              |
|------------------------------------------------------------------------------|----------------------|--------------|---------------|--------------|
|                                                                              | Total                | Non MNEs     | Domestic MNEs | Foreign MNEs |
| <i>(a) Effects of transfer pricing (no transfer pricing vs. no shifting)</i> |                      |              |               |              |
| North America                                                                | -0.16                | 0.36         | -0.85         | 0.35         |
| Europe                                                                       | -0.17                | -0.15        | -0.11         | -0.31        |
| Low tax                                                                      | -0.25                | -0.72        | 1.10          | -0.56        |
| Rest of world                                                                | -0.18                | -0.18        | -0.15         | -0.31        |
| <i>(b) Effects of profit shifting (no shifting vs. baseline)</i>             |                      |              |               |              |
| North America                                                                | <b>0.08</b>          | <b>-0.00</b> | <b>0.15</b>   | <b>0.15</b>  |
| Europe                                                                       | 0.06                 | -0.02        | 0.17          | 0.04         |
| Low tax                                                                      | <b>-0.04</b>         | <b>-0.33</b> | <b>-0.29</b>  | <b>0.64</b>  |
| Rest of world                                                                | 0.04                 | -0.01        | 0.08          | 0.08         |

# OECD Reform Proposal: Model Implementation

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**OECD Pillar 1:** revenue-based profit allocation. algebra

- Based on countries' shares of the firm's overall global sales.
- Independent of whether the firm has a physical presence in these countries (non-MNE exporters).

**OECD Pillar 2:** global minimum corporate income tax.

- Top-up rule: if firm based in  $i$  reports profits in  $j$  where  $\tau_j < \underline{\tau}$  then these profits are taxed in  $i$  at rate  $\underline{\tau} - \tau_j$ .
- Additional revenue for  $i$  is then

$$\tilde{R}_i = \sum_{j=1}^I \int_{\Omega_i} \max((\underline{\tau} - \tau_j), 0) \pi_{ij}^{PS}(\omega) d\omega.$$

# OECD Reform Proposal: Macro Effects

| Region                                       | Lost profits<br>(benchmark = 1) | Corp. tax<br>rev. (% chg.) | Value added<br>(% chg.) | Tech. capital (% chg.) |             |                  |
|----------------------------------------------|---------------------------------|----------------------------|-------------------------|------------------------|-------------|------------------|
|                                              |                                 |                            |                         | Total                  | Non<br>MNEs | Domestic<br>MNEs |
| <i>(a) Pillar 1: Profit reallocation</i>     |                                 |                            |                         |                        |             |                  |
| North America                                | 0.60                            | 2.54                       | -0.13                   | <b>-0.40</b>           | <b>0.15</b> | <b>-0.80</b>     |
| Low tax                                      | 0.69                            | -11.40                     | -0.13                   | 0.79                   | 0.23        | 1.35             |
| <i>(b) Pillar 2: Global minimum tax rate</i> |                                 |                            |                         |                        |             |                  |
| North America                                | 0.37                            | 3.24                       | -0.06                   | <b>-0.15</b>           | <b>0.08</b> | <b>-0.31</b>     |
| Low tax                                      | 0.49                            | -9.70                      | 0.02                    | 0.32                   | 0.36        | 0.28             |
| <i>(c) Pillars 1 &amp; 2 together</i>        |                                 |                            |                         |                        |             |                  |
| North America                                | 0.23                            | 4.36                       | -0.17                   | <b>-0.48</b>           | <b>0.17</b> | <b>-0.94</b>     |
| Low tax                                      | 0.33                            | -16.46                     | -0.13                   | 1.00                   | 0.48        | 1.51             |

*Notes:* For the low-tax region, lost profits are negative in both the benchmark equilibrium and in the policy counterfactuals, i.e., profits are shifted inward to the low-tax region.

# OECD Reform Proposal: Macro Effects

| Region                                       | Lost profits<br>(benchmark = 1) | Corp. tax<br>rev. (% chg.) | Value added<br>(% chg.) | Tech. capital (% chg.) |             |                  |
|----------------------------------------------|---------------------------------|----------------------------|-------------------------|------------------------|-------------|------------------|
|                                              |                                 |                            |                         | Total                  | Non<br>MNEs | Domestic<br>MNEs |
| <i>(a) Pillar 1: Profit reallocation</i>     |                                 |                            |                         |                        |             |                  |
| North America                                | <b>0.60</b>                     | <b>2.54</b>                | <b>-0.13</b>            | -0.40                  | 0.15        | -0.80            |
| Low tax                                      | 0.69                            | -11.40                     | -0.13                   | 0.79                   | 0.23        | 1.35             |
| <i>(b) Pillar 2: Global minimum tax rate</i> |                                 |                            |                         |                        |             |                  |
| North America                                | <b>0.37</b>                     | <b>3.24</b>                | <b>-0.06</b>            | -0.15                  | 0.08        | -0.31            |
| Low tax                                      | 0.49                            | -9.70                      | 0.02                    | 0.32                   | 0.36        | 0.28             |
| <i>(c) Pillars 1 &amp; 2 together</i>        |                                 |                            |                         |                        |             |                  |
| North America                                | <b>0.23</b>                     | <b>4.36</b>                | <b>-0.17</b>            | -0.48                  | 0.17        | -0.94            |
| Low tax                                      | 0.33                            | -16.46                     | -0.13                   | 1.00                   | 0.48        | 1.51             |

*Notes:* For the low-tax region, lost profits are negative in both the benchmark equilibrium and in the policy counterfactuals, i.e., profits are shifted inward to the low-tax region.

# OECD Reform Proposal: VA decomposition

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| Region                                       | Value added (% chg.) |              |               |              |
|----------------------------------------------|----------------------|--------------|---------------|--------------|
|                                              | Total                | Non MNEs     | Domestic MNEs | Foreign MNEs |
| <i>(a) Pillar 1: Profit reallocation</i>     |                      |              |               |              |
| North America                                | <b>-0.13</b>         | <b>-0.01</b> | <b>-0.30</b>  | <b>-0.05</b> |
| Low tax                                      | -0.13                | -0.10        | 0.36          | -0.56        |
| <i>(b) Pillar 2: Global minimum tax rate</i> |                      |              |               |              |
| North America                                | <b>-0.06</b>         | <b>0.01</b>  | <b>-0.10</b>  | <b>-0.13</b> |
| Low tax                                      | 0.02                 | 0.23         | 0.19          | -0.46        |
| <i>(c) Pillars 1 &amp; 2 together</i>        |                      |              |               |              |
| North America                                | <b>-0.17</b>         | <b>-0.02</b> | <b>-0.36</b>  | <b>-0.11</b> |
| Low tax                                      | -0.13                | 0.07         | 0.50          | -0.98        |



# OECD Reform Proposal: Macro Effects

| Region                                       | Lost profits<br>(benchmark = 1) | Corp. tax<br>rev. (% chg.) | Value added<br>(% chg.) | Tech. capital (% chg.) |             |                  |
|----------------------------------------------|---------------------------------|----------------------------|-------------------------|------------------------|-------------|------------------|
|                                              |                                 |                            |                         | Total                  | Non<br>MNEs | Domestic<br>MNEs |
| <i>(a) Pillar 1: Profit reallocation</i>     |                                 |                            |                         |                        |             |                  |
| North America                                | 0.60                            | 2.54                       | -0.13                   | -0.40                  | 0.15        | -0.80            |
| Low tax                                      | 0.69                            | -11.40                     | -0.13                   | <b>0.79</b>            | <b>0.23</b> | <b>1.35</b>      |
| <i>(b) Pillar 2: Global minimum tax rate</i> |                                 |                            |                         |                        |             |                  |
| North America                                | 0.37                            | 3.24                       | -0.06                   | -0.15                  | 0.08        | -0.31            |
| Low tax                                      | 0.49                            | -9.70                      | 0.02                    | <b>0.32</b>            | <b>0.36</b> | <b>0.28</b>      |
| <i>(c) Pillars 1 &amp; 2 together</i>        |                                 |                            |                         |                        |             |                  |
| North America                                | 0.23                            | 4.36                       | -0.17                   | -0.48                  | 0.17        | -0.94            |
| Low tax                                      | 0.33                            | -16.46                     | -0.13                   | <b>1.00</b>            | <b>0.48</b> | <b>1.51</b>      |

Notes: For the low-tax region, lost profits are negative in both the benchmark equilibrium and in the policy counterfactuals, i.e., profits are shifted inward to the low-tax region.

# OECD Reform Proposal: Macro Effects

| Region                                       | Lost profits<br>(benchmark = 1) | Corp. tax<br>rev. (% chg.) | Value added<br>(% chg.) | Tech. capital (% chg.) |             |                  |
|----------------------------------------------|---------------------------------|----------------------------|-------------------------|------------------------|-------------|------------------|
|                                              |                                 |                            |                         | Total                  | Non<br>MNEs | Domestic<br>MNEs |
| <i>(a) Pillar 1: Profit reallocation</i>     |                                 |                            |                         |                        |             |                  |
| North America                                | 0.60                            | 2.54                       | -0.13                   | -0.40                  | 0.15        | -0.80            |
| Low tax                                      | <b>0.69</b>                     | <b>-11.40</b>              | <b>-0.13</b>            | 0.79                   | 0.23        | 1.35             |
| <i>(b) Pillar 2: Global minimum tax rate</i> |                                 |                            |                         |                        |             |                  |
| North America                                | 0.37                            | 3.24                       | -0.06                   | -0.15                  | 0.08        | -0.31            |
| Low tax                                      | <b>0.49</b>                     | <b>-9.70</b>               | <b>0.02</b>             | 0.32                   | 0.36        | 0.28             |
| <i>(c) Pillars 1 &amp; 2 together</i>        |                                 |                            |                         |                        |             |                  |
| North America                                | 0.23                            | 4.36                       | -0.17                   | -0.48                  | 0.17        | -0.94            |
| Low tax                                      | <b>0.33</b>                     | <b>-16.46</b>              | <b>-0.13</b>            | 1.00                   | 0.48        | 1.51             |

*Notes:* For the low-tax region, lost profits are negative in both the benchmark equilibrium and in the policy counterfactuals, i.e., profits are shifted inward to the low-tax region.

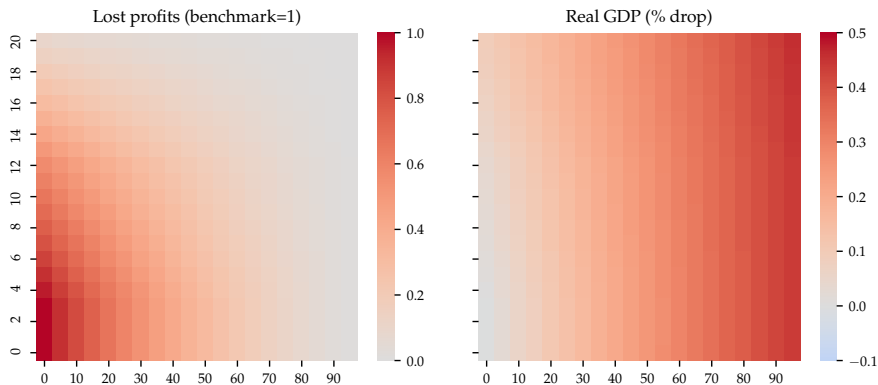
# OECD Reform Proposal: VA decomposition

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| Region                                       | Value added (% chg.) |              |               |              |
|----------------------------------------------|----------------------|--------------|---------------|--------------|
|                                              | Total                | Non MNEs     | Domestic MNEs | Foreign MNEs |
| <i>(a) Pillar 1: Profit reallocation</i>     |                      |              |               |              |
| North America                                | -0.13                | -0.01        | -0.30         | -0.05        |
| Low tax                                      | <b>-0.13</b>         | <b>-0.10</b> | <b>0.36</b>   | <b>-0.56</b> |
| <i>(b) Pillar 2: Global minimum tax rate</i> |                      |              |               |              |
| North America                                | -0.06                | 0.01         | -0.10         | -0.13        |
| Low tax                                      | <b>0.02</b>          | <b>0.23</b>  | <b>0.19</b>   | <b>-0.46</b> |
| <i>(c) Pillars 1 &amp; 2 together</i>        |                      |              |               |              |
| North America                                | -0.17                | -0.02        | -0.36         | -0.11        |
| Low tax                                      | <b>-0.13</b>         | <b>0.07</b>  | <b>0.50</b>   | <b>-0.98</b> |

# Varying the Sizes of the Pillars: North America

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*Notes:* X-axis in each plot represents the reallocation share for pillar 1. Y-axis in each plot represents the global minimum corporate income tax rate for pillar 2.

# Main Takeaways

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1. We develop **a model of international profit shifting** in which MNEs can transfer ownership of intangible capital to low-tax countries.
2. The key **economic trade-off**: profit shifting erodes high-tax countries' tax bases, but incentivizes the MNEs to invest in the intangible capital.
3. The landmark OECD reform has **sizeable, adverse macroeconomic effects** despite very small number of firms being engaged in profit shifting.
4. **Similar magnitude** to welfare effects of major trade liberalizations
  - U.S. gained 0.06% from NAFTA (**Caliendo and Parro, 2014**)
  - OECD gained 0.15% from China trade (**di Giovanni et al., 2014**)

# Calibration Overview

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| Parameter                        | Description                    | Value(s) | Target/source                            |
|----------------------------------|--------------------------------|----------|------------------------------------------|
| <i>(a) Assigned parameters</i>   |                                |          |                                          |
| $\varrho$                        | EoS between products           | 5        | Standard                                 |
| $N_j$                            | Population                     | Varies   | World Development Indicators             |
| $\tau_j$                         | Corporate income tax rate      | Varies   | Tørsløv, Wier, and Zucman (2022)         |
| <i>(b) Calibrated parameters</i> |                                |          |                                          |
| $\phi$                           | Technology capital share       | 0.11     | MNEs' intangible income share            |
| $A_i$                            | Total factor productivity      | Varies   | Real GDP                                 |
| $\eta_i$                         | Productivity dispersion        | Varies   | Large firms' employment share            |
| $\psi_i$                         | Utility weight on leisure      | Varies   | $L_i = N_i/3$                            |
| $\xi_{ij}$                       | Variable export cost           | Varies   | Bilateral imports/GDP                    |
| $\kappa_i^X$                     | Fixed export cost              | Varies   | Pct. of firms that export                |
| $\sigma_i$                       | Variable FDI cost              | Varies   | Foreign MNEs' share of value added       |
| $\kappa_i^F$                     | Fixed FDI cost                 | Varies   | Avg. emp. of firms w/ foreign affiliates |
| $\psi_{iLT}$                     | Cost of shifting profits to LT | Varies   | Total lost profits                       |
| $\psi_{iTH}$                     | Cost of shifting profits to TH | Varies   | Share of profits shifted to TH           |
| $\kappa_i^{TH}$                  | Fixed cost of TH affiliate     | Varies   | Avg. emp. of firms w/ TH affiliates      |

# Calibration: Region-specific target moments

---

| Region                                    | North America | Europe | Low-tax | RoW    | Tax haven |
|-------------------------------------------|---------------|--------|---------|--------|-----------|
| <i>(a) Region-specific target moments</i> |               |        |         |        |           |
| Population (NA = 100)                     | 100           | 92     | 11      | 1,323  | –         |
| Real GDP (NA = 100)                       | 100           | 80.78  | 14.57   | 297.10 | –         |
| Corporate tax rate (%)                    | 22.5          | 17.3   | 11.4    | 17.4   | 3.3       |
| Foreign MNEs' VA share (%)                | 11.12         | 19.82  | 28.73   | 9.55   | –         |
| Total lost profits (\$B)                  | 143           | 216    | –       | 257    | –         |
| Lost profits to TH (%)                    | 66.4          | 44.5   | –       | 71.1   | –         |
| Imports from... (% GDP)                   |               |        |         |        |           |
| North America                             | –             | 1.28   | 1.77    | 1.74   | –         |
| Europe                                    | 1.70          | –      | 12.39   | 3.78   | –         |
| Low tax                                   | 0.35          | 2.98   | –       | 0.59   | –         |
| Row                                       | 6.15          | 7.96   | 6.78    | –      | –         |

[return](#)

# Calibration: Internally-calibrated parameter values

| Region                                            | North America | Europe | Low-tax | RoW    | Tax haven |
|---------------------------------------------------|---------------|--------|---------|--------|-----------|
| <i>(b) Internally-calibrated parameter values</i> |               |        |         |        |           |
| TFP ( $A_i$ )                                     | 1.00          | 0.89   | 1.58    | 0.20   | —         |
| Prod. dispersion ( $\eta_i$ )                     | 4.28          | 4.31   | 4.83    | 4.12   | —         |
| Utility weight on leisure ( $\psi_i$ )            | 1.06          | 1.08   | 1.09    | 1.06   | —         |
| Fixed export cost ( $\kappa_i^X$ )                | 1.7e-3        | 3.5e-3 | 1.0e-3  | 1.4e-2 | —         |
| Variable FDI cost ( $\sigma_i$ )                  | 0.47          | 0.56   | 0.52    | 0.53   | —         |
| Fixed FDI cost ( $\kappa_i^F$ )                   | 1.80          | 1.59   | 0.46    | 8.75   | —         |
| Cost of shifting profits to LT ( $\psi_{iLT}$ )   | 3.40          | 0.38   | —       | 2.35   | —         |
| Cost of shifting profits to TH ( $\psi_{iTH}$ )   | 2.25          | 1.25   | —       | 1.76   | —         |
| Fixed FDI cost to TH ( $\kappa_i^{TH}$ )          | 0.09          | 0.06   | —       | 0.59   | —         |
| Variable trade cost from...                       |               |        |         |        |           |
| North America                                     | —             | 3.21   | 3.41    | 2.07   | —         |
| Europe                                            | 1.89          | —      | 1.69    | 1.33   | —         |
| Low tax                                           | 2.04          | 1.59   | —       | 1.56   | —         |
| RoW                                               | 2.26          | 2.59   | 3.01    | —      | —         |

[return](#)



# The Missing Profits of Nations

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Tørsløv, Thomas R and Wier, Ludvig S and Zucman, Gabriel (2020)  
computes missing profits by country:

- The thought experiment is that 1) absent profit shifting and 2) keeping total global profits fixed, how much more profit should be booked in non tax-haven countries.

The computation is done in two steps:

1. Computes excess profits in each tax haven:
  - Compare profit-to-wage ratio of foreign-owned affiliates and local firms in tax havens.
  - Purge out capital intensity differences; obtain total shifted profits of a tax haven from the profit-to-wage ratio gap.
2. Reallocated excess profits to non tax haven countries to:
  - Source countries, based on bilateral excess-risk services exports.
  - Parent countries, based on ownership.

# Example: Apple return

Before 2015: Profits shifted to Ireland via cross-licensing scheme

*...95 percent of Apple's R&D, the engine behind the success of Apple products, is conducted in the United States...[Apple Ireland] paid approximately \$5 billion to [Apple USA] as its share of the R&D costs. Over that same time period, [Apple Ireland] received profits of \$74 billion. The difference between [Apple Ireland's] costs and the profits, almost \$70 billion, is how much taxable income, in the absence of [Apple USA's] cost-sharing agreement with its own subsidiaries and its use of other tax loopholes, would otherwise have flowed to the United States. - Sen. Carl Levin, 2013*

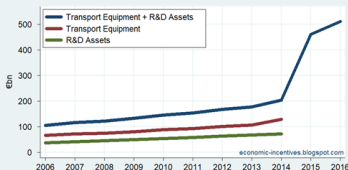
After 2015: Law changes forced Apple to sell IP to Irish affiliate, increasing Ireland's aggregate capital stock by 40% in one year

Gross External Debt: Non-IFSC FDI  
Direct investment debt - Debt liabilities to direct investors, €billion



Source: CSO: Balance of Payments

Capital Stock of Fixed Assets  
Gross Capital Stock of Transport Equipment and R&D Assets in 2015 prices



Source: Central Statistics Office

# Ireland's Service Trade [return](#)

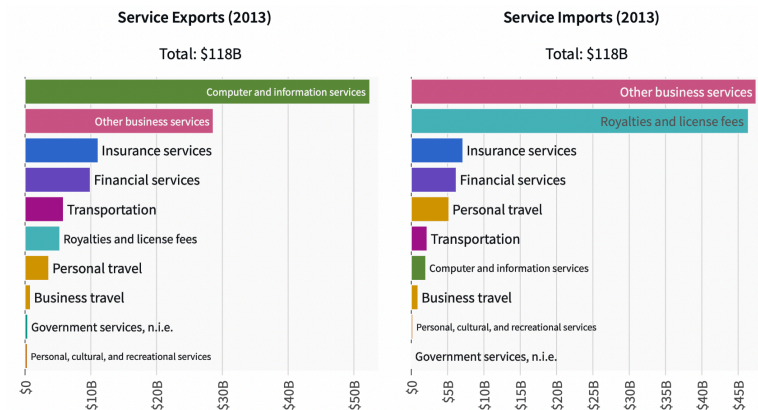


Figure: Ireland's service trade in 2013

# Consumer's Problem

---

Consumers choose labor supply  $L$  and consumption  $C$ :

$$U(C_i, L_i) = \max_{C_i, L_i} \left[ \log \left( \frac{C_i}{N_i} \right) + \psi \log \left( 1 - \frac{L_i}{N_i} \right) \right]$$

s.t.

$$P_i C_i = W_i L_i + (1 - \tau_i) D_i$$

return

# Final Goods Producer

---

The final goods producer of region  $i$  combines intermediate goods with a CES technology:

$$Q_j = \left[ \sum_{i=1}^J \int_{\Omega_{ji}} q_{ji}(\omega)^{\frac{\varrho-1}{\varrho}} d\omega \right]^{\frac{\varrho}{\varrho-1}}$$

- $\Omega_{ji}$ : the set of goods from  $i$  available in  $j$ .
- $q_{ji}$ : quantity of inputs
- $\varrho$ : elas. of sub. between varieties

Demand curves:

$$p_{ji}(\omega) = P_i Q_i^{\frac{1}{\varrho}} q_{ji}(\omega)^{-\frac{1}{\varrho}}, \quad (1)$$

The price index is :

$$P_j = \left[ \sum_{i=1}^J \int_{\Omega_{ji}} p_{ji}(\omega)^{1-\varrho} d\omega \right]^{\frac{1}{1-\varrho}}$$

# Technology

---

Technology of firm  $\omega$  in region

$$y_j(\omega) = \sigma_{ij} A_j a(\omega) (N_j z(\omega))^\gamma \ell_j(\omega)^\phi. \quad (2)$$

where

- $\sigma_{ij}$  is openness of  $j$  to FDI from  $i$
- $A_j$  is TFP in region  $j$
- $a$  is the firm-specific productivity
- $N_j$  is population in region  $j$
- $z$  is firm's intangible capital
- $\ell_j$  is labor hired in  $j$
- $\gamma$  and  $\phi$  are returns to scale parameters

return

- Firms from region  $i$  can serve the domestic market freely.
- Two options for serving foreign markets:
  - Export domestically produced goods. Fixed cost:  $\kappa_{ijX}$
  - Open a foreign affiliate and produce locally. Fixed cost:  $\kappa_{ijF}$
- The firm's resource constraints

$$y_i = q_{ii} + \sum_{j \in J_X} \xi_{ij} q_{ij}^X \quad (3)$$

$$y_j = q_{ij}, \quad j \in J_F \quad (4)$$

where

- $J_X \subseteq J \setminus i$  : set of foreign destinations to which the firm exports
- $J_F \subseteq J \setminus i$  : set of foreign destinations in which the firm operates a subsidiary

# Scale Choice

---

We use non-exporting foreign affiliate as an example.

Given  $z$ , an affiliate of firm  $\omega \in \Omega_i$  in region  $j$  chooses labor input  $l$  to maximize profit:

$$\begin{aligned}\pi_{ij}^F(a, z) &= \max_{q, \ell} p_{ij}(q)q - W_i \ell \\ &= \max_{\ell} P_j Q_j^{\frac{1}{\varrho}} (\sigma_{ij} A_j a)^{\frac{\varrho-1}{\varrho}} (N_j z)^{\gamma \frac{\varrho-1}{\varrho}} \ell^{\phi \frac{\varrho-1}{\varrho}} - W_j \ell\end{aligned}$$

From the FOC,  $\ell$  can be solved as:

$$\ell = \left\{ \left[ \frac{\phi(\varrho-1)}{\varrho} \right]^{\varrho} (P_j/W_j)^{\varrho} Q_j (\sigma_{ij} A_j a)^{\varrho-1} (N_j z)^{\gamma(\varrho-1)} \right\}^{\frac{1}{\phi + \varrho - \phi\varrho}}$$

return



# IP Choice

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R&D technology: number of workers required to produce 1 unit of intangible capital in country  $j$  is  $B_j$

Under free transferability, the optimal choice of  $z$  is

$$z = \left\{ \left( \frac{\phi + \varrho - \phi\varrho}{\gamma(\varrho - 1)} \right) \left[ \frac{(1 - \tau_i) W_i / A_i}{(1 - \tau_i) (\bar{R}_{ii} - \bar{C}_{ii}) + \sum_{j \in J_F} (1 - \tau_j) (\bar{R}_{ij} - \bar{C}_{ij})} \right] \right\}^{\frac{\phi + \varrho - \phi\varrho}{\gamma\varrho + \phi\varrho - \gamma - \phi - \varrho}}$$

Within the square bracket (the exponent outside is negative):

- The numerator is the marginal cost of producing  $z$ .
- The denominator is the marginal benefit.
- Adding transfer pricing and profit shifting will change optimal  $z$  through the denominator.

return

# Profit Shifting Choice

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From the FOC, optimal  $\lambda$  can be solved as (independent of  $z$ ):

$$\lambda = (C')^{-1} \left[ (1 - \varphi) \frac{(\tau_i - \tau_{i^*})}{1 - \tau_i} \right]$$

We can see that  $\lambda$ :

- decreases with the discount factor  $\varphi$ .
- decreases with lowest tax rate  $\tau_{i^*}$ .

return

# Firm's problem: free transfer of $z$

return

$$d_i^{FT}(\omega) = \max_{z, \ell, J_X, J_F, q} \left\{ (1 - \tau_i) \overbrace{\left[ p_{ii}(q_{ii})q_{ii} + \sum_{j \in J_X} (p_{ij}^X(q_{ij}^X)q_{ij}^X - W_i \kappa_{ijX}) - W_i(\ell_i + z/A_i) - W_i \sum_{j \in J_F} \kappa_{ijF} \right]}^{\text{Domestic parent profits}} \right. \\ \left. + \sum_{j \in J_F} (1 - \tau_j) \underbrace{[p_{ij}(q_{ij})q_{ij} - W_j \ell_j]}_{\text{Foreign subsidiary profits}} \right\} \quad (5)$$

subject to (1), (2), (3), and (4).

Simplify the notation:

$$\pi_i^D(a, z, J_X) = \max_{q_{ii}, \{q_{ij}^X\}_{j \in J_X}, \ell_i} \left\{ p_{ii}(q_{ii})q_{ii} + \sum_{j \in J_X} p_{ij}(q_{ij}^X)q_{ij}^X - W_i \ell_i \right\} \\ \text{s.t. } q_{ii} + \sum_{j \in J_X} \xi_{ij} q_{ij} = y_i = A_i a (N_i z)^\gamma \ell_i^\phi$$

and

$$\pi_{ij}^F(a, z) = \max_{q_{ij}, \ell_j} p_{ij}(q_{ij})q_{ij} - W_j \ell_j.$$

Thus, the conglomerate's problem can be written more succinctly as

$$d_i^{FT}(\omega) = \left\{ (1 - \tau_i) \left[ \pi_i^D(a, z; J_X) - W_i \left( z/A_i + \sum_{j \in J_X} \kappa_{ijX} + \sum_{j \in J_F} \kappa_{ijF} \right) \right] \right. \\ \left. + \sum_{j \in J_F} (1 - \tau_j) \pi_{ij}^F(a, z) \right\}$$

# Firm's Problem: transfer pricing

[return](#)

Building upon  $d^{FT}(a)$ , the TP version of the problem can be written as

$$d_i^{TP}(\omega) = \max_{z, J_X, J_F} \left\{ (1 - \tau_i) \left[ \pi_i^D(a, z; J_X) - W_i \left( z/A_i + \sum_{J \in J_X} \kappa_{ijX} + \sum_{j \in J_F} \kappa_{ijF} \right) + \overbrace{\sum_{j \in J_F} \vartheta_{ij}(z)z}^{\text{Licensing fees}} \right] \right. \\ \left. + \sum_{j \in J_F} (1 - \tau_j) \left[ \pi_{ij}^F(a, z) - \underbrace{\vartheta_{ij}(z)z}_{\text{Licensing fee}} \right] \right\}$$

# Firm's Problem: profit shifting

return

$$\begin{aligned}
 d_i^{PS}(\omega) = & \max_{z, J_X, J_F, \lambda_{LT}, \lambda_{TH}} \left\{ (1 - \tau_i) \left[ \pi_i^D(a, z; J_X) - W_i \left( z/A_i + \sum_{J \in J_X} \kappa_{ijX} + \sum_{j \in J_F} \kappa_{ijF} \right) \right. \right. \\
 & + \underbrace{\sum_{j \in J_F} (1 - \lambda_{LT} - \lambda_{TH}) \vartheta_{ij}(z) z}_{\text{Licensing fee receipts}} + \underbrace{(\varphi_i \lambda_{LT} + \varphi_i \lambda_{TH}) v_i(z) z}_{\text{Proceeds from selling } z} \\
 & - \underbrace{(\lambda_{LT} + \lambda_{TH}) \vartheta_{ii}(z) z}_{\text{Licensing fee payments}} - \underbrace{W_i \kappa_{iTH} 1(\lambda_{TH} > 0)}_{\text{Tax haven affiliate cost}} - \underbrace{C(\lambda_{TH} + C(\lambda_{LT})) \nu_i(z) z}_{\text{Cost of shifting } z} \Big] \\
 & + (1 - \tau_{LT}) 1_{(LT \in J_F)} \left[ \pi_{i,LT}^F(a, z) + \underbrace{\sum_{j \in J_F \cup \{i\} \setminus \{LT\}} \lambda_{LT} \vartheta_{ij}(z) z}_{\text{Licensing fee receipts}} - \underbrace{\varphi_i \lambda_{LT} v_i(z) z}_{\text{Cost of buying } z} - \underbrace{\vartheta_{iLT}(z) z}_{\text{Licensing fee pay}} \right] \\
 & + (1 - \tau_{TH}) 1_{(\lambda_{TH} > 0)} \left[ \underbrace{\sum_{j \in J_F \cup \{i\}} \lambda_{TH} \vartheta_{ij}(z) z}_{\text{Licensing fee receipts}} - \underbrace{\varphi_i \lambda_{TH} v_i(z) z}_{\text{Cost of buying } z} \right] \\
 & + \sum_{j \in J_F \setminus \{LT\}} (1 - \tau_j) \left[ \pi_{ij}^F(a, z) - \underbrace{\vartheta_{ij}(z) z}_{\text{Licensing fee}} \right] \Big\}
 \end{aligned}$$

Nominal GDP:

$$GDP_i = \sum_{j=1}^I \int_{\omega \in \Omega_j, i \in J_F(\omega)} p_{ji}(\omega) y_{ji}(\omega) d\omega.$$

Goods Trade:

$$EX_i^G = \sum_{j \neq i} \int_{\Omega_i} p_{ij}^X(\omega) (1 + \xi_{ij}) q_{ij}^X(\omega) d\omega,$$

$$IM_i^G = \sum_{j \neq i} \int_{\Omega_j} p_{ji}^X(\omega) (1 + \xi_{ji}) q_{ji}^X(\omega) d\omega.$$

## Services Trade:

- high-tax regions

$$EX_i^S = \sum_{j \neq i} \int_{\Omega_i} [1 - \lambda_{LT}(\omega) - \lambda_{TH}(\omega)] \vartheta_{ij}(\omega) z(\omega) d\omega$$

$$IM_i^S = \sum_{j \neq i} \int_{\Omega_i} [\lambda_{LT}(\omega) + \lambda_{TH}(\omega)] \vartheta_{ij}(\omega) z(\omega) d\omega + \sum_{j \neq i} \int_{\Omega_j} \vartheta_{ji}(\omega) z(\omega) d\omega$$

- low-tax regions:

$$EX_{LT}^S = \sum_{j \neq i} \int_{\Omega_i} [1 - \lambda_{TH}(\omega)] \vartheta_{ij}(\omega) z(\omega) d\omega + \sum_{j \neq i} \int_{\Omega_j} \lambda_{LT} \vartheta_{ji}(\omega) z(\omega) d\omega$$

$$IM_{LT}^S = \sum_{j \neq i} \int_{\Omega_i} \lambda_{TH}(\omega) \vartheta_{ij}(\omega) z(\omega) d\omega + \sum_{j \neq i} \int_{\Omega_j} [1 - \lambda_{LT}(\omega)] \vartheta_{ji}(\omega) z(\omega) d\omega$$

- tax haven:

$$EX_{TH}^S = \sum_{j=1}^I \int_{\Omega_j} \lambda_{TH} \vartheta_{ji}(\omega) z(\omega) d\omega$$



Net factor receipts and payments:

$$NFR_i = \sum_{j \neq i} \int_{\Omega_i} (1 - \tau_j) \pi_{ij}^{PS}(\omega) d\omega$$

$$NFP_i = \sum_{j \neq i} \int_{\Omega_j} (1 - \tau_i) \pi_{ji}^{PS}(\omega) d\omega$$

Labor market:

$$\begin{aligned}
 L_i = & \underbrace{\sum_{j=1}^I \int_{\Omega_j} \ell_{ji}(\omega) \, d\omega}_{\text{goods production}} + \underbrace{\int_{\Omega_i} z(\omega) / A_i \, d\omega}_{z \text{ production}} + \underbrace{\int_{\Omega_i} \left( \sum_{j \in J_X(\omega)} \kappa_i^X + \sum_{j \in J_F(\omega)} \kappa_i^F + \lambda_{TH}(\omega) > 0 \kappa_i^{TH} \right) d\omega}_{\text{fixed costs}} \\
 & + \underbrace{\int_{\Omega_i} (\mathcal{C}_{i,TH}(\lambda_{TH}) + \mathcal{C}_{i,LT}(\lambda_{LT})) \nu(\omega) z(\omega) \, d\omega}_{\text{costs of shifting } z}
 \end{aligned}$$

Government Budget Constraint:

$$T_i = \tau_i \sum_{j=1}^I \int_{\Omega_j} \pi_{ji}^{PS}(\omega) \, d\omega.$$

Balance of Payments:

$$EX_i^G + EX_i^S - IM_i^G - IM_i^S + NFR_i - NFP_i = 0.$$

# Pillar One: Residual profit

[return](#)

Under pillar 1, the tax base of a subsidiary in jurisdiction  $j$  is

$$T_j = \Pi_j^r + (1 - s) \cdot \Pi_j^R + s \cdot \frac{p_j y_j}{\sum_k p_k y_k} \cdot \Pi^R$$

where routine profit is

$$\Pi_j^r = \mu p_j y_j$$

Residual profit in jurisdiction  $j$  is

$$\Pi_j^R = \pi_j^{PS} - \mu p_j y_j$$

and total global residual profit

$$\Pi^R = \sum_i \Pi_i^R$$

Hence the problem of the MNE with profit reallocation is

$$\max_{z, J_x, J_F, \lambda_{LT}, \lambda_{TH}} \left\{ \sum_j \left( \pi_j^{PS} - \tau_j T_j \right) - W_i \left( \sum_{j \in J_X} \kappa_{ijX} - \sum_{j \in J_F} \kappa_{ijF} + \kappa_{iTH} 1(\lambda_{TH} > 0) \right) \right\}$$

We can show that the tax base  $T_{ij}$  can be derived in the model as:

$$T_{ii}(a, z) = (1 - \theta) \cdot \pi_i^D(a, z, J_X) + \theta \cdot \frac{R_{ii}(a, z)}{\sum_{k \in \{i\} \cup J_X \cup J_X} R_{ik}(a, z)} \cdot \sum_j \pi_i^D(a, z, J_X)$$

$$T_{ij}(a, z) = (1 - \theta) \cdot \pi_{ij}^F(a, z) + \theta \cdot \frac{R_{ij}(a, z)}{\sum_{k \in \{i\} \cup J_X \cup J_X} R_{ik}(a, z)} \cdot \sum_j \pi_j^F(a, z)$$